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Lab Task No: 4

Question 1: A* - First Three Expansions

Rule used: f(n) = g(n) + h(n), where g is the real cost so far, h is the guessed cost to the goal, and f is the two added together.

Step	Node	g	h	f
1	S	0	5	5
2	A	1	4	5
3	C	3	2	5

Working:
S: 0 + 5 = 5
A: 1 + 4 = 5
C: 3 + 2 = 5
G: 6 + 0 = 6

Frontier (nodes waiting to be expanded):
After S: A (5), B (8)
After A: C (5), B (8)
After C: G (6), B (8)

Final path: S - A - C - G **Total cost:** 6

Question 2: A* vs UCS

```
import heapq

graph = {
    'S': [('A', 1), ('B', 4)],
    'A': [('C', 2)],
    'B': [('G', 5)],
    'C': [('G', 3)],
    'G': []
}

h = {'S': 5, 'A': 4, 'B': 4, 'C': 2, 'G': 0}


def search(heuristic):
    queue = [(heuristic['S'], 0, 'S', ['S'])]
    best = {'S': 0}
    count = 0
    order = []

    while queue:
        f, g, node, path = heapq.heappop(queue)
        if g != best[node]:
            continue
        order.append(node)
        if node == 'G':
            return path, g, order, count
        count += 1
        for child, cost in graph[node]:
            new_g = g + cost
            if new_g < best.get(child, float('inf')):
```

```
                best[child] = new_g
                heapq.heappush(queue, (new_g + heuristic[child], new_g, child, path + [child]))
    return [], 0, order, count


print("A*:", search(h))
zero_h = {node: 0 for node in graph}
print("UCS:", search(zero_h))
```

Results:

Detail	A*	UCS
Visit order	S, A, C, G	S, A, C, B, G
Path	S-A-C-G	S-A-C-G
Cost	6	6
Expansions	3	4

Both find the same best path with the same cost. A* does less work (3 expansions instead of 4) because the h values guide it away from B.

Question 3: Heuristic Overestimate

Only one change: h(C) goes from 2 to 10. The real remaining cost from C to G is 3, so 10 is an overestimate.

Step	Node	g	h	f
1	S	0	5	5
2	A	1	4	5
3	B	4	4	8
4	G	9	0	9

C now has f = 3 + 10 = 13, so A* skips it and goes through B instead. **New path:** S - B - G with **cost 9**, which is worse than the best cost of 6. So overestimating h makes A* give a wrong (non-best) answer.

GitHub Link

https://github.com/umair-900/Ai-lab-Task